

BIOINFORMATICS MINI-PROJECT 2

Project 2 — Comparative Structural Analysis

Hen Egg-White Lysozyme: PDB 1FDL vs PDB 1AKI

Student End Report
Beginner-level research-profile project

1. Abstract

This project compared hen egg-white lysozyme in two structural contexts: antibody-bound lysozyme in **1FDL** and isolated lysozyme in **1AKI**. The goal was to determine whether the overall fold remains broadly similar when the protein is free versus bound to an antibody. The comparison used sequence landmarks, database information and visual inspection in Mol*. It is a qualitative comparison, not a numerical structural-alignment study.

2. Scientific Question

Does hen egg-white lysozyme retain a broadly similar 3D fold when it is alone compared with when it is bound to an antibody?

3. Structures Compared

Feature	1FDL	1AKI
Context	Lysozyme bound to Fab D1.3	Isolated lysozyme
Method	X-ray diffraction	X-ray diffraction
Resolution	2.50 Å	1.50 Å
Chain	C (author Y)	A
Length	129 residues	129 residues
Organism	Gallus gallus	Gallus gallus

4. Methodology

I opened both PDB entries in Mol*, inspected the lysozyme chain in each, checked sequence landmarks, viewed the 3D shapes, and compared the reported experimental resolutions.

5. Sequence Check

The 1AKI lysozyme sequence began with **KVFGRCELAA**, matching the beginning seen for the lysozyme in 1FDL. The terminal landmark **VQAWIRGCRL** was also observed. Both records report a 129-residue lysozyme chain.

6. Visual 3D Comparison

The overall lysozyme shape appeared broadly similar in the free structure and the antibody-bound structure. The fold remained recognizable even though 1FDL includes the antibody complex. This is a visual observation, not a claim that every atom occupies the same position.

7. Resolution

1AKI has a reported resolution of 1.50 Å, compared with 2.50 Å for 1FDL. A smaller Å value generally corresponds to finer structural detail supported by the experiment.

8. Interpretation

The main beginner-level observation is that lysozyme retains a broadly similar overall fold in the two structures. A structural alignment and a metric such as RMSD would be required to measure the difference precisely.

9. What I Learned

I learned how to compare two PDB structures of the same protein, use sequence information as a verification step, interpret resolution, and distinguish visual comparison from quantitative structural analysis.

10. Limitations and Future Work

No RMSD, TM-score or automated structural alignment was calculated. Future work should align 1AKI with the lysozyme chain from 1FDL, calculate RMSD, and inspect residues with the largest positional differences.

11. Conclusion

The two structures show a broadly similar overall lysozyme fold despite their different biological contexts. The comparison provides a useful foundation for a later quantitative structural-alignment exercise.

12. References

RCSB PDB 1FDL: <https://www.rcsb.org/structure/1FDL>

NCBI Structure 1AKI: <https://www.ncbi.nlm.nih.gov/Structure/pdb/1AKI>

IBDC 1AKI summary: <https://ibdc.dbt.gov.in/isda/summary/1AKI/>

Appendix — Project Screenshots

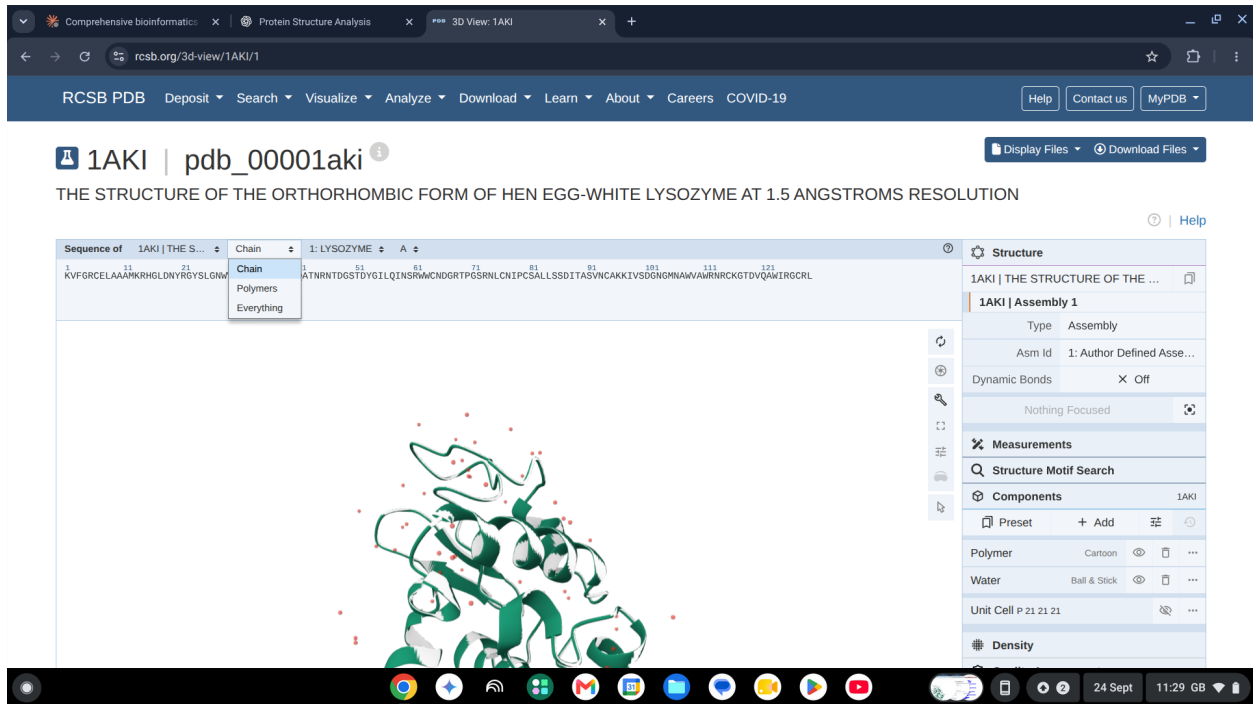


Figure A1. Mol* view of PDB 1AKI.

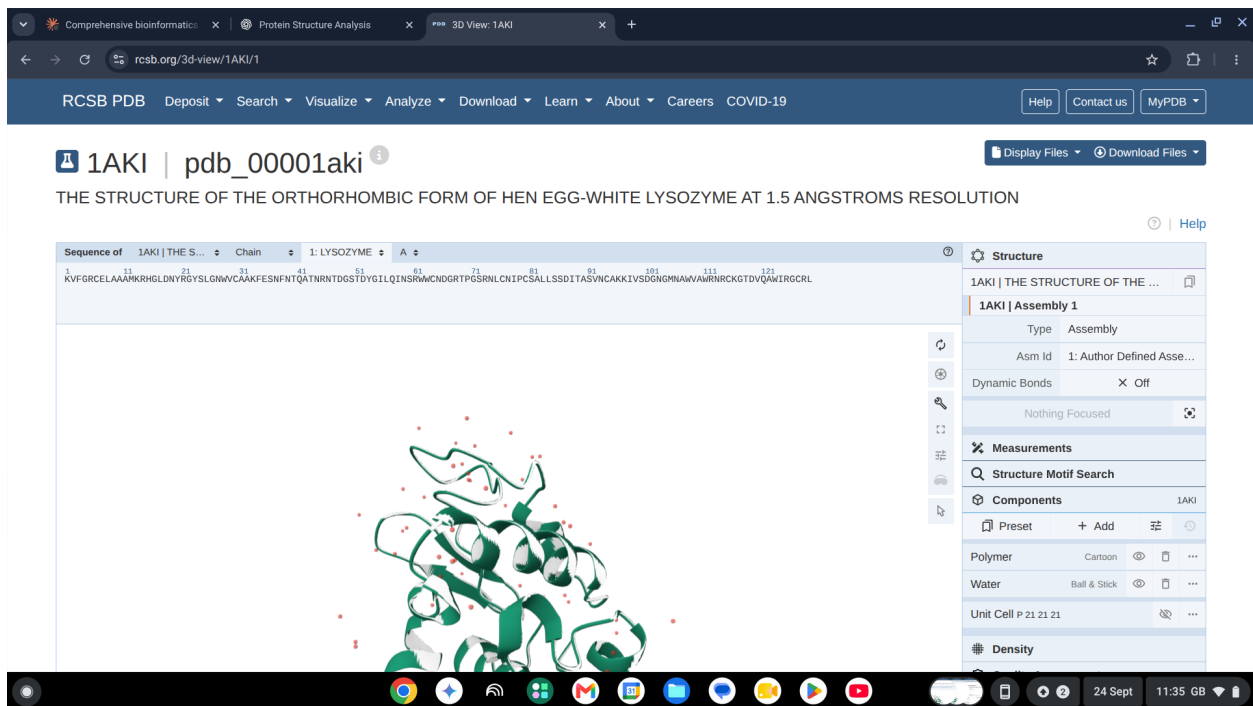


Figure A2. Additional 1AKI Mol* view.

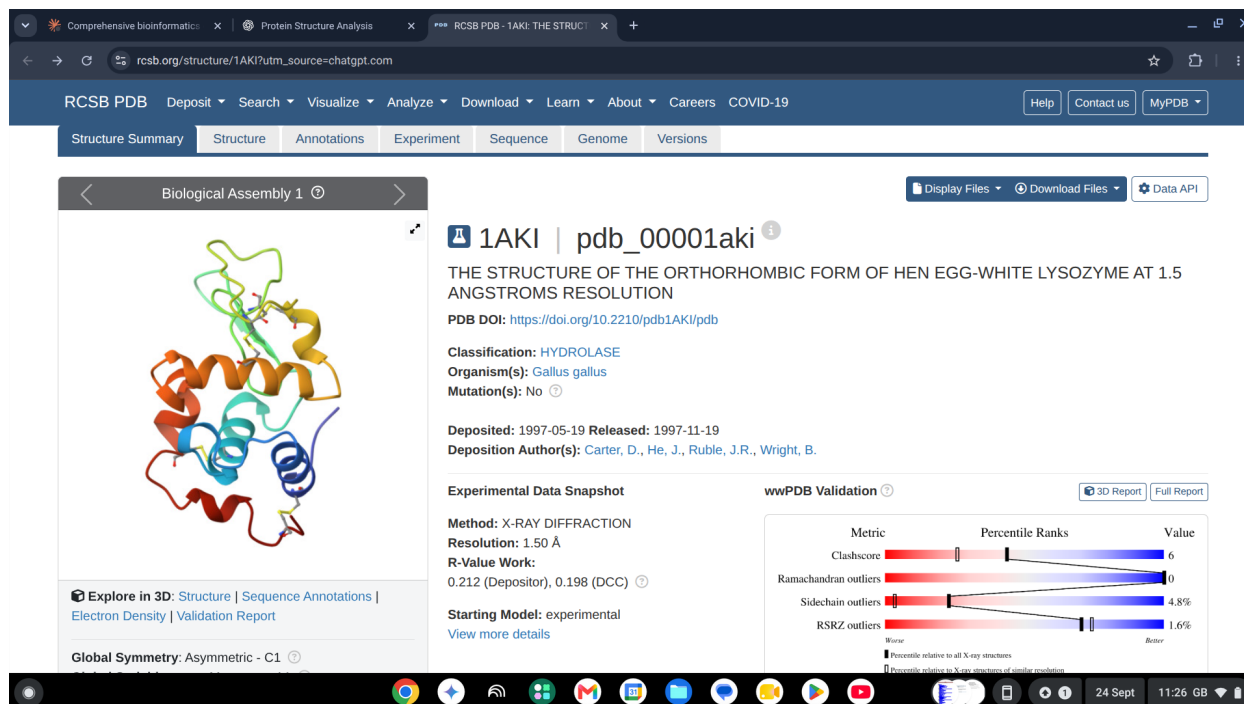


Figure A3. RCSB PDB overview for 1AKI.

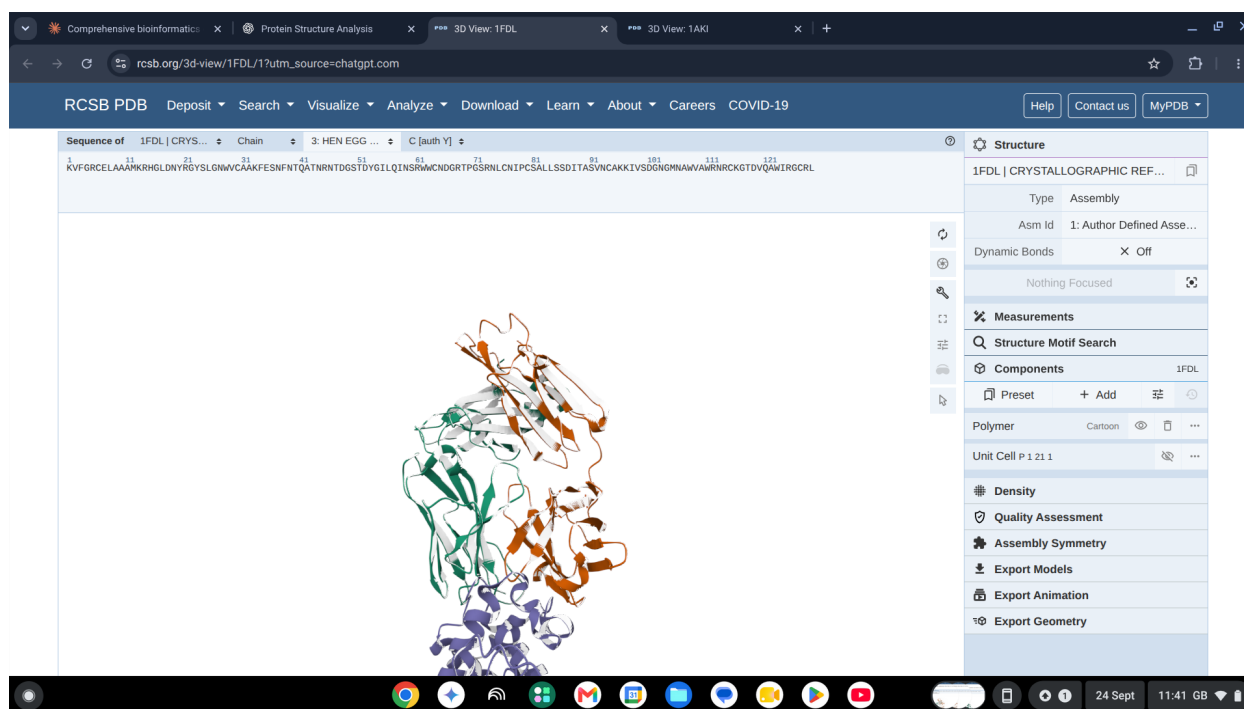


Figure A4. 1FDL Mol* view used for comparison.